

Lab 10: Gene Regulation & Epigenetics

BIOL-1

Name: _____ Date: _____

Overview

All cells in your body have the same DNA, yet a neuron looks and acts very differently from a skin cell. This is due to **gene regulation**: the ability to turn specific genes on or off. In this lab, you will explore two mechanisms of regulation: the *lac* operon (in bacteria) and epigenetic modification (in eukaryotes).

Learning Objectives

By the end of this lab, you will be able to:

1. Model the components of the *lac* operon and explain inducible gene regulation.
2. Demonstrate how repressors block transcription.
3. Simulate epigenetic modifications (methylation, acetylation) using physical models.
4. Explain how environmental factors (diet, stress) can influence gene expression.

Activity 1: The Lac Operon Simulation

Materials:

- DNA strip (Promoter, Operator, Structural Genes)
- Repressor Protein model (e.g., clay/foam)
- Lactose Molecule model
- RNA Polymerase model

Procedure:

1. **Condition A: No Lactose present.**
 - Place the **Repressor** on the **Operator**.
 - Try to move **RNA Polymerase** from the **Promoter** to the genes.
 - **Question:** Can RNA Polymerase transcribe the genes? Why or why not?

2. Condition B: Lactose added.

- Attach the **Lactose** molecule to the **Repressor**.
- Observe: Does the Repressor stay on the DNA or fall off?
- Now try to move **RNA Polymerase**. Can it transcribe the genes?

1. Why is this system beneficial for the bacterium? Why not make the enzymes all the time?

Activity 2: Epigenetics — Chromatin Modeling

Materials:

- Yarn/string (DNA)
- Beads/spools (Histones)
- Tape (Methyl/Acetyl groups)

Procedure:

1. **Heterochromatin (Off):** Tightly wind the yarn around the beads. Try to "read" the sequence written on the yarn. Is it accessible?
2. **Euchromatin (On):** Loosen the winding. Add "Acetyl groups" (tape) to the histones to keep them spaced apart.
 - Now try to read the sequence. Is it accessible?

2. How does adding acetyl groups (acetylation) to histones generally affect gene expression?

3. What environmental factors might cause your cells to change their epigenetic markers (methylation/acetylation)? Give one example.

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Conclusion

Gene regulation allows cells to be efficient and specialized. Operons control prokaryotic genes in response to the environment, while chromatin structure controls eukaryotic access to DNA.